Notes: Ketcham, Lucarelli & Powers (AER, 2015)
Paying Attention or Paying Too Much in Medicare Part D

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1 Big picture

- **Question.** Does having many Medicare Part D prescription drug plans make consumers more inert and less responsive to potential savings, or do consumers learn and switch when their status quo becomes expensive?
- **Setting.** Medicare Part D created a large, subsidized market in which beneficiaries choose among private prescription drug plans (PDPs). The paper studies nonpoor PDP enrollees from 2006–2010 using a random 20 percent CMS sample.
- **Main descriptive result.** By 2010, about half of beneficiaries who initially chose a PDP were no longer in their original plan, so inertia is not absolute.
- **Main switching result.** Individuals are more likely to switch when staying in their status quo plan would generate higher above-minimum out-of-pocket spending.
- **Main choice-set result.** Larger choice sets do not generally increase inertia. Adding lower-cost plans increases switching, while adding only very expensive plans has little effect.
- **Main time/experience result.** Switching overall and responsiveness to status quo costs do not decline over time. In fact, responsiveness to large cost increases is often greater in later years.
- **Welfare implication.** Switching saves money. Among those who switch, mean savings are typically over $100 per year, and cumulative savings from switching exceed $1 billion for the 2006–2010 cohort.
- **Takeaway.** The evidence gives little support to the view that more options or more time in the market mechanically heighten inertia in Part D. Consumers respond to financially relevant plan differences when those differences are salient and large.

2 Incremental contribution of the paper

- **From initial choice mistakes to dynamic choice.** Earlier Part D work, especially Abaluck and Gruber, focused on whether consumers chose low-cost plans at market entry. This paper studies whether consumers subsequently learn, switch, and respond to cost changes over multiple years.
- **Direct test of choice overload in a high-stakes field setting.** Rather than inferring choice overload from high above-minimum spending, the paper directly estimates how within-person changes in the number of available plans affect switching.
- **Distinguishes quantity from quality of options.** The paper's key innovation is to classify plans by their individual-specific relative cost. This separates adding more low-cost options from adding more irrelevant high-cost options.
- **Uses administrative CMS data and institutional cost calculators.** The authors build accurate plan-by-person counterfactual out-of-pocket costs using internal CMS data, plan design details, prices, formularies, premiums, and claims.
- **Separates plan turnover from plan crosswalk noise.** The paper constructs a refined crosswalk to distinguish true switching from plan termination, consolidation, and contract-ID changes.
- **Tests both experience and time.** It evaluates whether consumers become more inert as they age within the market or as the market matures, and finds little evidence for worsening inertia.
- **Policy contribution.** The results caution against simply reducing the number of Part D plans. What matters is whether additional plans are financially relevant to consumers.

Interpretation: The incremental contribution is not merely a larger sample. It is the combination of longitudinal enrollee histories, person-specific counterfactual plan costs, and a decomposition of choice-set size by relative cost.

3 Data and measurement

3.1 Study population and administrative data

The paper uses a random 20 percent sample of Medicare beneficiaries from CMS. The main switching analysis restricts to individuals who are enrolled in a PDP for two consecutive years, receive no low-income subsidy, are at least age 65, are alive at the end of the year, and are enrolled for the full year in one PDP. The sample includes millions of switching decisions from 2007–2010.

- **Enrollment data:** identify plan enrollment and plan switching.
- **Plan files:** provide premiums, plan characteristics, formularies, pharmacy networks, and pricing.
- **Prescription Drug Event data:** provide the universe of drug claims used to calculate plan-specific out-of-pocket costs.
- **Health controls:** include medical spending and chronic condition indicators from CMS files.

Interpretation: These data are unusually rich because they observe both actual choices and what each individual would have paid in many unchosen plans.

3.2 Cost calculator and spending measures

For individual i, plan j, and year t, let

$$C_{ijt} = \text{annual premium}_{jt} + \text{calculated out-of-pocket drug spending}_{ijt}.$$

Actual plan spending is denoted C_{it}, and the minimum available plan cost is

$$\min_j C_{ijt}.$$

The realized above-minimum spending is

$$AboveMinimum_{it} = C_{it} - \min_j C_{ijt}.$$

For switching regressions, the key status quo relative cost is

$$A_{it} = C_{ij_{t-1}t} - \min_j C_{ijt},$$

where j_{t-1} is the plan chosen in the prior year.

Interpretation: A_{it} is the amount the beneficiary could save in year t by switching from the status quo to the current minimum-cost plan. It is the central economic incentive for switching.

3.3 Alternative cost models

The paper uses three approaches to calculate counterfactual costs:

- **Ex post approach:** applies current-year drug utilization to all available plans.
- **Ex ante approach:** mimics the CMS plan finder, using information available before the next year's plan choice.
- **Rational expectations approach:** approximates expected future spending based on similar individuals' costs.

Interpretation: The robustness of results across these approaches is important because counterfactual spending in unchosen plans is inherently unobserved.

3.4 Choice-set measures

The paper uses two types of choice-set measures:

- total stand-alone plans available in the region-year;
- number of plans within individual-specific cost bins relative to the minimum-cost plan: within \$100, \$100–200, \$200–300, \$300–400, \$400–500, and more than \$500.

Interpretation: This distinction is central. A larger total menu can be unhelpful if the new plans are expensive, but valuable if it adds low-cost alternatives.

4 Models and empirical specifications

4.1 Baseline switching model

The baseline linear probability model is

$$S_{it} = \theta_t + \theta_i + \alpha_1 A_{it} + \alpha_2 N_{it} + \alpha_3 H_{it} + u_{it}.$$

Here:

- $S_{it} = 100$ if individual i switches PDPs for year t;
- θ_t are year fixed effects;
- θ_i are individual fixed effects;
- A_{it} is above-minimum spending from staying in the status quo plan;
- N_{it} is either the total number of plans or the vector of plan counts by relative-cost bins;
- H_{it} includes time-varying health controls.

Interpretation: Individual fixed effects use within-person variation, reducing concerns that inherently attentive consumers both face different menus and switch more.

4.2 Choice-overload test

To test whether larger menus make consumers less responsive to savings, the model adds an interaction:

$$S_{it} = \theta_t + \theta_i + \alpha_1 A_{it} + \alpha_2 N_{it} + \alpha_3 H_{it} + \alpha_4 (A_{it} \times N_{it}) + u_{it}.$$

Interpretation: If choice overload makes people less responsive, α_4 should be negative: as the menu grows, the incentive to switch away from an expensive status quo should weaken.

4.3 Time and experience model

To test whether people become more or less responsive over time and experience, the model adds interactions:

$$S_{it} = \theta_t + \theta_i + \alpha_1 A_{it} + \alpha_2 A_{it} \times \theta_t + \alpha_3 H_{it} + \alpha_4 A_{it} \times N_{it} + \alpha_5 A_{it} \times E_{it} + \alpha_6 A_{it} \times H_{it} + u_{it}.$$

Interpretation: Time effects capture how the market changes from 2007 to 2010; experience captures how the same person changes as they spend more time in the Part D market.

5 Identification and empirical design

5.1 Core identification logic

The paper exploits within-person changes over time in three objects: the cost of staying in the status quo plan, the number and cost distribution of alternative plans, and the beneficiary's experience in the market. Individual fixed effects absorb time-invariant preferences for switching, health level, cognitive skill, and risk aversion.

Interpretation: The key comparison is not one person with many plans versus a different person with fewer plans. It is the same person facing a different status quo cost and a different choice set in another year.

5.2 Why within-person variation matters

Beneficiaries differ in attention, ability, trust in insurers, cognitive capacity, and health complexity. If these traits are fixed or slow moving, individual fixed effects remove them. The identifying variation comes from plan and formulary changes, premium changes, regional menu changes, and evolving drug needs.

Interpretation: This design is much stronger than a cross-sectional comparison of plan-count regions, though it still depends on the assumption that time-varying unobservables are adequately controlled.

5.3 Choice overload versus economic relevance

A naive choice-overload test would regress switching on the total number of plans. The paper argues that this is incomplete because many added plans are dominated or irrelevant. The key empirical design therefore decomposes N_{it} into cost bins.

Interpretation: This is the conceptual core of the paper. More choices should not be treated as one homogeneous object. A new low-cost plan and a new very expensive plan have opposite economic meanings.

5.4 Main threats and responses

- **Measurement of counterfactual costs.** The true spending in unchosen plans is unobserved. The paper uses multiple cost models to test robustness.
- **Endogenous status quo.** Prior choices partly determine the current status quo. Individual fixed effects and the focus on changes in plan costs mitigate, but do not eliminate, this concern.
- **Health shocks.** Changes in health may change both drug needs and switching. The paper controls for time-varying health spending and diagnoses.
- **Plan terminations and forced switches.** The paper constructs crosswalks to distinguish true switching from administrative plan changes.
- **Generalizability.** The sample excludes low-income-subsidy recipients and focuses on early Part D years.

6 Tables and figures

The paper contains tables and no original figures. The visuals below are directly cropped from the paper. Related adjacent tables are placed on the same row where readable, following the requested compact layout.

6.1 Table 1: Share of People No Longer Enrolled in Their Original Plan; Table 2: Trends in Out-of-Pocket Spending

Read: Table 1 shows plan turnover. Among those starting in 2006, 53 percent were no longer in the original plan by 2010. Table 2 shows that total out-of-pocket spending and above-minimum spending decline from 2006 levels, with above-minimum spending remaining below 2006 and 2007 levels in 2010.

Economic message: The raw descriptive patterns already challenge extreme inertia: many beneficiaries leave original plans, and spending relative to the minimum does not deteriorate over time.

Table 1—Share of People No Longer Enrolled in Their Original Plan at the Beginning of Each Subsequent Year

	2007	2008	2009	2010
ong nonpoor PDP enrollees who started in				
006	18.8	32.8	44.0	53.0
007	—	26.2	42.6	53.8
008	—	—	24.8	42.1
009	—	—	—	33.0

(a) Table 1: share no longer enrolled in original plan.

Table 2—Trends in Out-of-Pocket Spending and Above-Minimum Spending among Nonpoor PDP Enrollees, 2006–2010

	2006	2007	2008	2009	2010
Total OOP spending on premiums and prescription drugs ($)	1,590	1,382	1,312	1,435	1,492
Year to year percent change in OOP spending	—	−13.1	−5.1	9.4	3.9
Percent change in OOP spending relative to 2006	—	−13.1	−17.5	−9.7	−6.2
Ex post approach with perfectly inelastic demand for drugs					
OOP Spending in minimum-cost plan ($)	1,070	1,036	1,019	1,104	1,163
Above-minimum OOP spending ($)	520	347	292	331	329
Above-minimum OOP spending as percent of total OOP (%)	33	25	22	23	22
Ex post approach with price elasticity of demand for drugs of −0.54					
OOP Spending in minimum-cost plan ($)	1,235	1,145	1,078	1,153	1,217
Above-minimum OOP spending ($)	355	237	234	282	275
Above-minimum OOP spending as percent of total OOP (%)	22	17	18	20	18

Note: OOP is out-of-pocket.

(b) Table 2: OOP and above-minimum spending trends.

6.2 Table 3: Switching and Savings from Switching; Table 4: Spending, Above-Minimum Spending, and Switching among the 2006–2010 Cohort

Read: Table 3 shows that annual switching is around 11 percent and that switchers save over $100 on average in each year. Table 4 restricts to a balanced 2006–2010 cohort and shows similar patterns.

Economic message: Switching is not random churn. It generates meaningful monetary savings, and those savings persist in the balanced cohort.

Table 3—Switching and Savings from Switching, 2007–2010

	2007	2008	2009	2010
Switched between PDPs from the prior year (%)	11.7	10.9	10.6	10.7
Ex post approach with perfectly inelastic demand for drugs				
Mean savings from switching ($)	178	256	254	206
Median savings from switching ($)	132	223	222	154
Percent that saved >$0 from switching (%)	66.1	81.7	86.2	77.4
Ex post approach with price elasticity of demand for drugs of −0.54				
Mean savings from switching ($)	102	194	195	144
Median savings from switching ($)	101	158	180	112
Percent that saved >$0 from switching	65.1	82.7	88.0	80.9

(a) Table 3: switching rates and annual savings from switching.

Table 4—Spending, Above-Minimum Spending, and Switching among the 2006–2010 Cohort

	2006	2007	2008	2009	2010
Switched PDPs from prior year (%)	—	11.4	11.6	9.6	11.8
Total OOP spending on premiums and prescription drugs ($)	1,519	1,504	1,438	1,597	1,691
OOP spending in minimum-cost plan ($)	1,006	1,138	1,115	1,238	1,331
Above-minimum OOP spending ($)	514	367	324	359	360
Above-minimum OOP spending as percent of total OOP (%)	33.8	24.4	22.5	22.5	21.3

Notes: OOP is out-of-pocket. The sample is restricted to the 306,315 that began in the PDP market without a low-income subsidy on January 1, 2006 and remained enrolled in a PDP without a low-income subsidy through December 31, 2010. Minimum and above-minimum spending are defined using the ex post approach and assuming

(b) Table 4: balanced-cohort spending and switching.

6.3 Table 5: Cumulative Switching and Savings from Switching; Table 6: Average Number of Prescription Drug Plans Available

Read: Table 5 shows cumulative switching and cumulative savings. By 2010, 27.6 percent of current enrollees had ever switched, generating more than $1 billion in cumulative savings under the inelastic approach. Table 6 shows that the average menu contains about 51 plans, with large within-person variation and substantial variation by relative cost category.

Economic message: Table 5 quantifies the value of switching. Table 6 shows why the paper's choice-set decomposition matters: some plans are near the minimum cost while many are far more expensive.

Table 5—Cumulative Switching and Savings from Switching, 2007–2010

	2007	2008	2009	2010
Percent of current enrollees that have ever previously switched between PDPs (%)	11.7	16.9	23.1	27.6
Actual OOP spending by those who ever previously switched ($)	1,882	1,480	1,500	1,540
Ex post approach with perfectly inelastic demand for drugs				
OOP spending if stayed in original plan and never switched ($)	2,062	1,687	1,725	1,773
Annual savings by current enrollees who ever previously switched ($)	180	207	225	233
Savings as a percent of total OOP spending (%)	9.6	14.0	15.0	15.1
Total non-subsidy PDP full year enrollees	7,462,806	7,743,191	7,790,845	7,787,562
Annual savings from cumulative switching ($)	157,306,398	271,298,803	405,055,147	500,802,537
Total cumulative savings from switching ($)	157,306,398	428,605,201	833,660,347	1,334,462,885
Ex post approach with price elasticity of demand for drugs of -0.54				
OOP spending if stayed in original plan and never switched ($)	1,987	1,650	1,686	1,734
Annual savings by current enrollees who ever previously switched ($)	105	170	186	194
Savings as a percent of total OOP spending (%)	5.6	11.5	12.4	12.6
Total non-subsidy PDP full year enrollees	7,462,806	7,743,191	7,790,845	7,787,562
Annual savings from cumulative switching ($)	91,820,275	222,880,629	334,867,424	416,977,220
Total cumulative savings from switching ($)	91,820,275	314,700,905	649,568,329	1,066,545,549

Notes: OOP is out-of-pocket.

(a) Table 5: cumulative switching and savings.

Table 6—Average Number of Prescription Drug Plans Available to the Study Sample, Overall and by Individual-Specific Relative Cost Categories

	Overall	2006	2007	2008	2009	2010	Average within-person Minimum	Average within-person Maximum
Total stand-alone plans available	51.1	43.1	55.8	54.5	50.2	47.1	46.3	55.3
Number of plans available within								
$100 of minimum	4.8	3.0	6.2	5.8	4.0	4.0	2.2	8.2
$100–200 of minimum	8.8	4.4	11.3	10.6	7.6	7.3	4.6	13.6
$200–300 of minimum	10.4	6.5	12.6	10.8	10.4	9.3	6.3	14.5
$300–400 of minimum	8.6	7.2	10.0	8.9	8.4	7.6	5.2	12.1
$400–500 of minimum	5.5	6.2	5.8	5.2	5.6	5.4	2.9	8.6
>$500 of minimum	13.0	16.0	9.8	13.2	14.3	13.5	7.6	19.4

Notes: The set of available prescription drug plans excludes employer-sponsored plans because those are not open to those not employed by the firm. The count by category of relative costs are from the ex post version assuming a price elasticity of demand for drugs of -0.54.

(b) Table 6: number of plans overall and by relative-cost category.

6.4 Table 7: Parameter Estimates from Models of Switching Prescription Drug Plans; Table 8: Alternative Categories of Number of Plans by Cost

Read: Table 7 shows that higher status quo above-minimum spending increases switching. More lower-cost alternatives also increase switching. Interactions between status quo costs and menu size do not support strong choice-overload effects. Table 8 shows the conclusion is robust to alternative cost-category groupings.

Economic message: The results distinguish economic relevance from menu size. More cheap alternatives encourage switching; more expensive alternatives do little.

Table 7—Parameter Estimates from Models of Switching Prescription Drug Plans, 2007–2010

	Number of plans overall				Number of plans by cost			
	No interactions		Interactions		No interactions		Interactions	
Above-minimum spending if stay in status quo plan ($100s)	2.88	[0.39]***	4.99	[2.16]**	4.03	[0.48]***	6.80	[2.04]***
Number of available plans	0.39	[0.31]	0.50	[0.34]				
Above-minimum spending if stay ($100s) × number of available plans			−0.04	[0.04]				
Number of available plans within								
$100 of minimum					0.60	[0.25]**	0.90	[0.31]***
$100–200 of minimum					0.61	[0.28]**	0.60	[0.31]*
$200–300 of minimum					0.54	[0.26]**	0.43	[0.30]
$300–400 of minimum					0.41	[0.24]*	0.32	[0.27]
$400–500 of minimum					0.43	[0.26]*	0.36	[0.31]
>$500 of minimum					0.00	[0.24]	0.58	[0.28]**
Above-minimum spending if stay ($100s) × number of available plans within								
$100 of minimum							−0.08	[0.05]
$100–200 of minimum							0.03	[0.04]
$200–300 of minimum							0.03	[0.04]
$300–400 of minimum							−0.01	[0.04]
$400–500 of minimum							−0.02	[0.04]
>$500 of minimum							−0.13	[0.04]***
2008	0.10	[1.50]	0.11	[1.49]	1.74	[1.55]	−0.08	[1.67]
2009	−0.20	[2.23]	−0.20	[2.24]	2.32	[2.06]	−0.02	[2.26]
2010	1.16	[3.42]	1.09	[3.45]	3.64	[2.94]	1.55	[3.18]

Notes: Observations: 3,645,570. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models also include individual fixed effects and controls for time-varying health. The number of plans by cost category and above-minimum spending in the status quo plan are defined using the ex post approach and assuming a price elasticity of demand for drugs of -0.54.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

(a) Table 7: baseline switching model and choice-overload interaction.

Table 8—Parameter Estimates from Models of Switching with Alternative Categories of the Number of Plans by Cost, 2007–2010

Model A. Number of available plans within		
$500 of minimum	0.51	[0.27]*
>$500 of minimum	−0.06	[0.25]
Model B. Number of available plans within		
$400 of minimum	0.54	[0.26]**
>$400 of minimum	0.09	[0.25]
Model C. Number of available plans within		
$300 of minimum	0.62	[0.26]**
>$300 of minimum	0.24	[0.24]

Notes: Observations: 3,645,570. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models also include individual fixed effects and controls for time-varying health. The number of plans by cost category and above-minimum spending in the status quo plan are defined using the ex post approach and assuming a price elasticity of demand for drugs of -0.54.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

(b) Table 8: robustness to alternative cost bins.

6.5 Table 9: Switching Models Allowing Differences over Time; Table 10: Conditional Within-Person Difference in Switching

Read: Table 9 interacts status quo costs with years, experience, age, health, and plan counts. It shows greater responsiveness to status quo costs in later years and lower responsiveness with age, Alzheimer's disease, and higher drug spending. Table 10 converts coefficients into conditional within-person differences relative to 2007.

Economic message: The evidence does not show worsening inertia over time. Instead, consumers become more responsive to large cost increases, while individual-specific experience has mixed effects.

Table 9—Parameter Estimates from Models of Switching that Allow for Differences over Time, 2007–2010

	Number of plans overall				Number of plans by cost	
	Full sample		Excluding early 2006 enrollees		Full sample	
Above-minimum spending if stay in status quo plan ($100s)	4.08	[3.88]	2.16	[4.10]	2.21	[3.72]
Above-minimum spending if stay ($100s) × 2008	3.21	[0.43]***	3.10	[0.45]***	3.36	[0.53]***
Above-minimum spending if stay ($100s) × 2009	3.85	[0.70]***	3.78	[0.65]***	4.62	[0.70]***
Above-minimum spending if stay ($100s) × 2010	3.79	[1.17]***	3.74	[1.13]***	4.86	[1.13]***
Above-minimum spending if stay ($100s) × months of experience	−0.06	[0.02]**	−0.06	[0.02]***	−0.07	[0.02]***
Above-minimum spending if stay ($100s) × age in 2010	−0.25	[0.05]***	−0.21	[0.06]***	−0.22	[0.04]***
Above-minimum spending if stay ($100s) × has Alzheimer's disease	−0.72	[0.11]***	−0.56	[0.15]***	−0.79	[0.09]***
Above-minimum spending if stay ($100s) × current gross drug spending ($1,000s)	−0.03	[0.01]***	0.00	[0.00]**	−0.02	[0.01]**
Current gross drug spending ($1,000s)	0.22	[0.06]***	0.15	[0.06]***	0.27	[0.05]***
Number of available plans	0.41	[0.34]	0.32	[0.39]		
Above-minimum spending if stay ($100s) × number of available plans	−0.02	[0.07]	0.01	[0.07]		
Number of available plans within						
$100 of minimum					0.63	[0.30]**
$100–200 of minimum					0.36	[0.30]
$200–300 of minimum					0.13	[0.29]
$300–400 of minimum					0.06	[0.26]
$400–500 of minimum					0.11	[0.29]
>$500 of minimum					0.33	[0.28]
Above-minimum spending if stay × number of available plans within						
$100 of minimum					0.01	[0.07]
$100–200 of minimum					0.11	[0.07]
$200–300 of minimum					0.12	[0.07]*
$300–400 of minimum					0.07	[0.06]
$400–500 of minimum					0.05	[0.06]
>$500 of minimum					−0.05	[0.07]
2008	−6.53	[1.75]***	−5.64	[1.88]***	−6.88	[1.77]***
2009	−7.33	[2.11]***	−7.31	[2.20]***	−8.68	[1.93]***
2010	−3.78	[3.36]	−2.79	[3.83]	−5.32	[2.96]*
Observations	3,645,570		2,077,059		3,645,570	

Notes: Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models include year fixed effects, individual fixed effects, and controls for time-varying health. The results in

(a) Table 9: switching responsiveness over time and by characteristics.

Table 10—Estimated Conditional Within-Person Difference in Switching between 2007 and Later Years in Response to an Increase in the Status Quo Plan's Costs

	Estimated difference from 2007 in the probability of switching					
	2008		2009		2010	
Increase in above-minimum spending if remain in the status quo plan						
$100	−4.04	[1.51]***	−4.93	[1.97]**	−2.16	[3.15]
$200	−1.55	[1.37]	−2.53	[1.95]	−0.55	[3.12]
$300	0.94	[1.37]	−0.13	[2.07]	1.07	[3.28]
$400	3.43	[1.49]**	2.27	[2.30]	2.69	[3.59]
$500	5.93	[1.72]***	4.67	[2.61]*	4.31	[4.02]
$1,000	18.38	[3.51]***	16.67	[4.74]***	12.40	[7.08]*

Notes: Estimates are based on the results from the first model provided in the preceding table. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets.
***Significant at the 1 percent level.
**Significant at the 5 percent level.

(b) Table 10: conditional difference in switching relative to 2007.

6.6 Table 11: Ex Post Inelastic Models; Table 12: Ex Ante Models

Read: Table 11 repeats the switching models under perfectly inelastic ex post demand. Table 12 uses ex ante plan-finder-style costs under elastic and inelastic assumptions. Both preserve the main findings: status quo cost increases switching, and lower-cost alternatives increase switching.

Economic message: The results are not driven by one particular cost-calculator assumption.

Table 11—Parameter Estimates from Ex Post Models of Switching Assuming Perfectly Inelastic Demand for Drugs, 2007–2010

	Number of plans overall				Number of plans by cost			
	No interactions		Interactions		No interactions		Interactions	
Above-minimum spending if stay in status quo plan ($100s)	1.86	[0.20]***	3.54	[1.37]**	2.08	[0.22]***	3.97	[1.36]***
Number of available plans	0.41	[0.28]	0.52	[0.29]*				
Above-minimum spending if stay ($100s) × number of available plans			−0.03	[0.03]				
Number of available plans within								
$100 of minimum					0.55	[0.26]**	0.88	[0.29]***
$100–200 of minimum					0.56	[0.29]*	0.61	[0.29]**
$200–300 of minimum					0.46	[0.27]*	0.40	[0.28]
$300–400 of minimum					0.42	[0.25]*	0.33	[0.27]
$400–500 of minimum					0.49	[0.26]*	0.45	[0.29]
>$500 of minimum					0.32	[0.26]	0.58	[0.28]**
Above-minimum spending if stay ($100s) × number of available plans within								
$100 of minimum							−0.08	[0.03]**
$100–200 of minimum							0.01	[0.03]
$200–300 of minimum							0.04	[0.03]
$300–400 of minimum							0.02	[0.03]
$400–500 of minimum							−0.01	[0.03]
>$500 of minimum							−0.06	[0.03]**
2008	1.18	[1.48]	1.12	[1.50]	1.76	[1.49]	1.30	[1.58]
2009	1.66	[2.05]	1.63	[2.06]	2.30	[2.00]	1.67	[2.11]
2010	2.91	[3.09]	2.87	[3.12]	3.71	[2.94]	3.23	[3.08]

Notes: Observations: 3,645,570. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models include year fixed effects, individual fixed effects and controls for time-varying health. The number of plans by cost category and above-minimum spending in the status quo plan are defined using the ex post approach and assuming perfectly inelastic demand for drugs.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

(a) Table 11: ex post models with perfectly inelastic demand.

Table 12—Parameter Estimates from Ex Ante Models of Switching, 2007–2010

	Elasticity of demand for drugs = −0.54				Perfectly inelastic demand for drugs			
	Number of plans overall		Number of plans by cost		Number of plans overall		Number of plans by cost	
Above-minimum spending if stay in status quo plan ($100s)	1.78	[0.35]***	2.34	[0.52]***	1.12	[0.16]***	1.26	[0.19]***
Number of available plans	0.41	[0.29]			0.42	[0.28]		
Number of available plans within								
$100 of minimum			0.50	[0.26]*			0.50	[0.26]*
$100–200 of minimum			0.55	[0.28]*			0.55	[0.29]*
$200–300 of minimum			0.48	[0.26]*			0.45	[0.27]*
$300–400 of minimum			0.41	[0.25]*			0.41	[0.25]
$400–500 of minimum			0.47	[0.27]*			0.47	[0.27]*
>$500 of minimum			0.18	[0.26]			0.37	[0.27]
2008	0.18	[1.41]	1.12	[1.41]	0.74	[1.40]	1.06	[1.40]
2009	0.58	[2.05]	1.87	[1.93]	1.66	[1.95]	1.95	[1.90]
2010	1.86	[3.13]	3.30	[2.85]	2.85	[2.95]	3.34	[2.84]

Notes: Observations: 3,644,298. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models include year fixed effects, individual fixed effects, and controls for time-varying health. The above-minimum spending in the status quo plan is defined using the ex ante approach.
***Significant at the 1 percent level.
**Significant at the 5 percent level.

(b) Table 12: ex ante switching models.

6.7 Table 13: Rational Expectations Models of Switching

Read: Table 13 uses a rational-expectations-style expected-cost measure. The status quo cost coefficient remains positive and large, and the cost-bin pattern remains similar.

Economic message: The core conclusions survive the strongest alternative approach to expected drug spending.

Table 13—Parameter Estimates from "Rational Expectations" Models of Switching, 2007–2010

	Elasticity of demand for drugs = −0.54				Perfectly inelastic demand for drugs			
	Number of plans overall		Number of plans by cost		Number of plans overall		Number of plans by cost	
Above-minimum spending if stay in status quo plan ($100s)	5.48	[0.54]***	6.40	[0.67]***	2.57	[0.39]***	2.76	[0.44]***
Number of available plans	0.32	[0.40]			0.44	[0.28]		
Number of available plans within								
$100 of minimum			0.54	[0.28]*			0.46	[0.29]
$100–200 of minimum			0.55	[0.28]*			0.57	[0.29]*
$200–300 of minimum			0.43	[0.27]			0.47	[0.25]*
$300–400 of minimum			0.31	[0.26]			0.38	[0.25]
$400–500 of minimum			0.27	[0.27]			0.46	[0.25]*
>$500 of minimum			−0.18	[0.28]			0.38	[0.26]
2008	−2.32	[1.67]	0.28	[1.67]	−0.74	[1.49]	0.40	[1.55]
2009	−3.86	[2.78]	−0.5	[2.27]	1.38	[2.12]	1.66	[2.01]
2010	−1.76	[4.45]	1.24	[3.28]	2.90	[3.22]	3.52	[2.99]

Notes: Observations: 3,483,504. Standard errors from cluster-robust variance estimator clustered at the region-year level are in brackets. All models include year fixed effects, individual fixed effects and controls for time-varying health. The number of plans by cost category and above-minimum spending in the status quo plan are defined using the "rational expectations" approach.
***Significant at the 1 percent level.

Table 13: rational-expectations switching models.

7 Short summary

- The paper studies dynamic PDP switching in Medicare Part D from 2006–2010.
- About half of initial enrollees are no longer in their original PDP by 2010.
- Switchers save meaningful amounts of money.
- Status quo above-minimum costs strongly predict switching.
- Total menu size does not support simple choice overload.
- The cost composition of the menu matters: low-cost alternatives increase switching, expensive alternatives do not.
- Responsiveness to large cost increases does not decline over time.
- Multiple cost models–ex post, ex ante, and rational expectations–produce the same qualitative conclusions.

8 Conclusion

8.1 Core conclusion

Ketcham, Lucarelli, and Powers show that Medicare Part D beneficiaries are not simply passive or increasingly inert. Many switch plans, switching saves money, and individuals are responsive to the financial cost of remaining in the status quo plan. The paper therefore provides little support for the view that large choice sets or longer time in the market mechanically heighten inertia.

8.2 Economic intuition

The intuition is that not all options impose equal cognitive costs or provide equal choice value. A menu with many dominated or expensive plans may be confusing and irrelevant, but a menu with additional low-cost plans provides meaningful opportunities. Consumers appear more likely to switch when the menu contains financially relevant alternatives.

8.3 Policy implication

The results caution against policies that mechanically reduce the number of Part D plans. A cap that removes low-cost alternatives could make consumers worse off. A better policy implication is to improve comparison tools, highlight low-cost alternatives, and reduce irrelevant complexity without eliminating valuable options.

8.4 Incremental takeaway

The paper's incremental contribution is to move the Part D choice debate from a static question–did people choose poorly at entry?–to a dynamic question–do people respond, switch, and learn as costs and options evolve? Its answer is more optimistic about consumer attention than many earlier choice-overload interpretations.

8.5 Limitations

The study covers early Part D years and excludes low-income-subsidy recipients. It also relies on modeled counterfactual spending in unchosen plans. The authors address this through multiple cost models, but true preferences over plans may include non-cost attributes such as pharmacy convenience, brand preference, risk protection, and hassle costs.

8.6 Final takeaway

In Medicare Part D, paying attention and paying too much are both real possibilities. But the evidence shows that many beneficiaries do pay attention when the dollar stakes are large, and more options are harmful only when the added options are not financially relevant.